



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End-Spring Semester Examination 2023-24

Date of Examination: _____ Session: (FN/AN) _____ Duration: 3 hrs.
Full Marks : 50 Subject No.: NA20202 Subject : Marine Hydrodynamics
Department/Center/School: Ocean Engineering and Naval Architecture
Specific charts, graph paper, log book etc., required
Special Instructions (if any): NO

Please note that the notations and variables used in the question paper are same as that of used in the class and hence not explained here.

Answer any five

1. The hydrodynamic force can be given in the form of

$F_i = -\rho \iint_{S_b} \left[\frac{\partial \phi_j}{\partial t} + \frac{1}{2} \nabla \phi_j \cdot \nabla \phi_j \right] n_i dS$. Show that, it can also be expressed in the form of $F_i = -\rho \frac{\partial}{\partial t} \iint_{S_b} [\phi_j \cdot n_i] dS + \rho \iint_{S_b} \left[\left(\frac{\partial \phi_j}{\partial n} \right) \nabla \phi_j - \frac{1}{2} \nabla \phi_j \cdot \nabla \phi_j \cdot n_i \right] dS$ after further simplification, if the force term is expressed in the form of $F_i(t) = -\dot{U}_j m_{ij}$, then define m_{ij} , write the expression for m_{ij} , and show that $m_{ij} = m_{ji}$, 4+1+2+3

2. Show that $F(x, y, z, t) = 0$ will be a boundary surface if $\frac{DF}{Dt} = 0$. From that, derive the kinematic free surface boundary condition. Also derive dynamic free surface condition. Discuss how we can modify these two condition under the assumption of linearity. 3+3+2+2

3. Suppose the wave elevation of a 2-dimensional plane progressive wave is defined as $z = \eta(x, t)$, where $\eta = A \cos(kx - \omega t)$. Then show that velocity potential for that wave may be represented as $\phi = \frac{Ag}{\omega} \frac{\cosh k(z+h)}{\cosh(kh)} \sin(kx - \omega t)$. Derive that water particle trajectory for the deep water condition. 4+6

4. Derive the expression for the total pressure in deep water condition. Show that under the assumption of the linearity, the total pressure is zero at the crest, but it is not zero at trough. Discuss the reason for that. 10

5. Write the expression for dispersion relation for finite water depth. What is the physical significance of dispersion relation? What is phase velocity? What is group velocity? show that for deep water case, phase velocity is twice the group velocity. If phase velocity of a plane progressive wave is $4m/s$. What would be the corresponding wave number if the wave is travelling in deep water? Prove that all frequency wave travel at same velocity in shallow water region. 1+2+1+1+2+2+1

6. Show that, if a 2-dimensional body generates a wave with an amplitude A because of its steady motion, then the associated drag force is $\frac{1}{4}\rho g A^2$. Write the 6 DoF equation of motion and define all the terms present in that equation. 7+3

----- All the best -----